Offline build: UNIROUTE_PIPELINE.md S1–S12

S1 Ingest & clean
last user msg + ≤ 2 turns · redact secrets · language tag · de-duplicate

S2 Split by conversation (never by turn)

S_tr train
everything else (≥ 5,000)
unlabeled

S_val validation
400 labeled

S_cal
100 human items ⊂ S_val

S_te test
300 labeled
locked until S11

S3 Model registry · cost per model (normalized 0–1) · anchor (pairwise) · holdout models (experiment mode)

S4 Embed all prompts
local multilingual embedder

S5 K-means on S_tr
K ∈ {1} ∪ {3 … N_val/50}
≥ 50 val prompts per cluster

S6 Generate answers on S_val ∪ S_te
every model, once, cached

Pilot (§P) 4 models · 200 val · 150 test
judge **both** ways · humans label S_cal
criteria: A cross-eval · B human agreement
C stability · D separability · tie → pointwise

decides method

S7 Judge (chosen method)
PT: score 1–10, pass ≥ τ
PW: vs frozen anchor, both orders

Judge–human agreement ≥ 75%?

no

Revise rubric / τ

yes

clusters

S8 Fingerprints Ψ from S_val only
PT: fail rate, smoothing α = 2
PW: Bradley–Terry P(lose to anchor)

S9 Choose K: split-validation on S_val (5 halvings), max deferral area

S10 Choose λ on S_val: quality target or cost budget

S11 Evaluate on S_te (first and only use)
UniRoute vs always-best · always-cheapest · ZeroRouter · K-NN
deferral area · QNC · similarity to production · experiment mode

S12 Router artifact (versioned)
method · embedder · K, centroids, names · Ψ · ĉ · λ · hashes

API calls only in S6 and S7 (orange). Everything else is free and can be re-run.
Golden rules: test never used for any choice · every call cached · answers shared by both methods · train split unlabeled · fingerprints from validation only · seeds recorded